

Emergent heterogeneity in compartmental disease models and its relation to super-spreading

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1 Introduction

Real outbreaks have significant heterogeneity in the number of secondary cases – including *superspreading*, marked by a small fraction of individuals causing a disproportionately large fraction of realized transmission [12, 18]. Superspreading can help explain why epidemics take off in some times and places but not others, despite apparently similar contexts [11, 4]; and leveraging knowledge of superspreading can make control efforts more effective [5, 14, 17].

Superspreading is salient for disease dynamics and control strategies at the beginning of outbreaks [6]. Correspondingly, theoretical treatment of superspreading emphasizes individual-level transmission patterns, often while assuming a fully susceptible population [12, 9]. Theoretical analysis of superspreading has focused on models that link observed secondary case counts to a continuous random variable, the *individual reproductive number*, via a Poisson sampling process. Canonically, the individual reproductive number is modeled as a gamma distribution, giving rise to negative binomial secondary case distributions [12]. In this way, variation from the stochastic process of successful transmission is conceptually separated from a more intrinsic individual-level heterogeneity [in expected infectiousness] that may emerge from a plethora of potential differences in biology, contact patterns, or behaviors between individuals [3, 1, 7].

Over longer timescales, the processes of susceptible depletion and replenishment, which are elegantly captured in mechanistic compartmental models, drive population-scale disease dynamics. Despite shared theory motivating models that highlight individual-level transmission, and those that emphasize population-level transmission dynamics, a gap persists in how each model framework treats individual-level transmission heterogeneity.

The canonical SIR (Susceptible-Infected-Removed) model is based on ordinary differential equations (ODEs). It assumes constant, uniform infectivity and recovery across all individuals during the infectious stage [8, 10]. Although

single-infective-class compartmental models seem homogeneous, even the canonical SIR model has two sources of *emergent heterogeneity*. The changing effective reproductive number arising from susceptible dynamics, drives variation between fractions of the population infected at different times; and the exponentially distributed recovery times assumed by the ODE, drive variation in the number of infections attributable to fractions of the population infected at a given time. In other words, even fully deterministic ODE models predict variation around basic reproductive number $\mathcal{R}_0$, the average number of secondary cases caused by a case in a fully susceptible population, in the expected secondary cases.

In practice, many epidemiological modelers also include *explicit heterogeneity*, for example by adding compartments for different demographic or geographic groups [16], which can have different average rates of mixing, transmission, or recovery. Explicit heterogeneity reproduces superspreading in linear compartmental models [13].

Here we delineate these different sources of individual variation in transmission inscribed in simple dynamical models. We trace how these sources compose into observed or observable heterogeneity in different settings, for example contrasting how heterogeneity emerges in dynamical models versus linearized, or branching-process models. Throughout, we highlight the understudied but critically important role of disease dynamics *per se* in generating individual-level heterogeneity. We aim to spur the development of theory and methods that better integrate heterogeneity and dynamics, ultimately facilitating feedback between data on incidence and individual-level transmission chains, mechanistic model specification, identification, validation, and intervention design.

Box 1. Linking variance.

During the early stages of an outbreak, individual-level secondary-cases counts are typically modeled using a negative-binomial distribution, with mean μ and shape, or “dispersion”, parameter k . This distribution is, implicitly or explicitly, linked to a Gamma distribution with the same mean and shape parameter values. Indeed, the Gamma distribution describes the variation among individuals in expected infectiousness.

We note a couple of points. First, the squared coefficient of variation of the underlying Gamma distribution is $\kappa = 1/k$, so that its variance $\sigma_{\text{act}}^2 = \kappa\mu^2$. The resulting variance in secondary-case counts is $\sigma^2 = \kappa\mu^2 + \mu$, with the last term arising from the Poisson step of the negative-binomial construction. Second, this relationship between the SCV of

The statistic k^{-1} increases with variability around the mean in both distributions. In particular, for the negative binomial distribution, k^{-1} is equal to the excess variance with respect to the Poisson distribution with mean μ , divided by μ^2 . For the Gamma distribution, k^{-1} equals the squared coefficient of variation. As k^{-1} approaches zero, the negative binomial distribution converges to the Poisson distribution with mean μ ,

while Gamma distribution converges to a point mass at μ .

We therefore advocate using $\kappa = k^{-1}$, as a (generalized) *dispersion* parameter, with larger values indicating greater heterogeneity in expected infectiousness and secondary cases.

2 Results

Demographic stochasticity can generate emergent heterogeneity even in the absence of explicit differences in individual-level rates. In outbreaks modeled using the linearized SIR compartmental models (with susceptible, infected, and removed compartments), this heterogeneity can be quantitatively characterized. In these models, the infectious period follows an exponential distribution, resulting in an exponentially distributed expected infectiousness (or *individual reproductive number*) with $\kappa = 1$. Consequently, the secondary-case distribution (or *offspring distribution*), modeled as a Poisson mixture of the expected infectiousness, follows a geometric distribution with dispersion κ (see Box).

[JD: Is that really what Box is doing, though? Or more about linking the emergent stochasticity in the deterministic vs. demographic-stochastic models?]

Panel (a) of Figure 1 shows patterns of emergent heterogeneity in secondary-case numbers during the early stage of an outbreak, when the population is fully susceptible. The distributions of both expected infectiousness and the number of secondary cases depend on $\mathcal{R}_0$, i.e., the average number of cases caused by a typical infector in a fully susceptible population.

Under a stochastic SIR model, where the number of secondary cases arises from Poisson draws from expected infectiousness, inequality in secondary cases—that is, the relationship between the fraction of new infections and the fraction of infectors—depends on the mean, whereas inequality in expected infectiousness does not (Figure 1, panel b). In fact, the share of Poisson noise in overall inequality shrinks as $\mathcal{R}_0$ increases, and the inequality in secondary cases approaches its theoretical limit, namely, the inequality in expected infectiousness (Figure 1, panel b). As $\mathcal{R}_0$ becomes smaller, an increasing fraction of infected individuals generate no secondary infections, so the individuals who do transmit make up a smaller fraction of all infectors. Indeed, the “20/80 rule”—that 80% of secondary infections arise from only 20% of infectious individuals—is attained when $\mathcal{R}_0$ falls just below 0.4. **[AA: don’t know why]** However, even when $\mathcal{R}_0$ exceeds the epidemic threshold, the distribution of secondary cases can still display substantial heterogeneity.

However, the susceptible population depletes in real outbreak. To what extent does the incorporation of changes in the susceptible pool affect these distributions? To address this, we simulated simple, stochastic SIR epidemics with different $\mathcal{R}_0$ values in a population of 10000 individuals. For each infected individual, we recorded the number of secondary cases they generated during the outbreak. Panel (a) of Figure 2 shows realized distributions of sec-

ondary cases caused by individual infectors over the entire outbreaks. Unlike in the linearized scenario, where the susceptible pool remains fully populated, the secondary case distributions, computed across the entire outbreak, are indistinguishable across a broad range of the key parameter $\mathcal{R}_0$ (compare panels a of Figure 1 and Figure 2). This result is unexpected. Faster and larger dynamical changes occur when $\mathcal{R}_0$ is large (Figure S1), which would typically be expected to correspond to larger variation in the expected infectiousness distribution and, hence, secondary-case distribution.

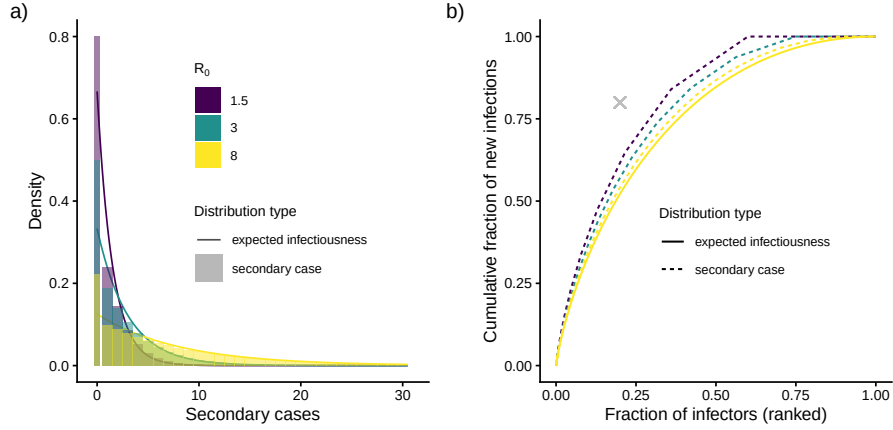


Figure 1: **Heterogeneity emerges even from a simple, linearized compartmental model** due to implicit variation in recovery times among infectors. Panel a: Both expected infectiousness (density curves) and secondary case distributions (density histograms) for the outset of an SIR epidemic, where the population is fully susceptible depend on $\mathcal{R}_0$. Panel b: Inequality curves, defined as the cumulative fraction of new infections versus the cumulative fraction of infectors, computed from the expected infectiousness across varying $\mathcal{R}_0$, are identical and visually indistinguishable due to overplotting. In contrast, inequality curves derived from the secondary case distribution exhibit differences. As $\mathcal{R}_0$ increases, the inequality observed in the secondary case distribution decreases and approaches the expected infectiousness inequality curve. The cross marks “20-80” point, where 20% of infectors cause 80% of infections. In panel a, because the first bin (at zero) sits at the boundary of support for each distribution, we have plotted this bin as double the density and half the width; this adjustment preserves area-to-area correspondence with the probability density function, while facilitating visual comparison of the heights of the density and mass functions.

To clarify these counterintuitive observations, we considered a standardized deterministic SIR model in which the time is measured in units of the mean infectious period. We examined how heterogeneity evolves in epidemiologically relevant cohorts, defined as groups infected simultaneously, over the course of

the outbreak. A large $\mathcal{R}_0$ initially generates significant expected infectiousness. Later, rapid depletion of susceptible individuals reduces the expected infectiousness. In contrast, a small $\mathcal{R}_0$ initially causes smaller expected infectiousness, but the slower susceptible exhaustion allows for considerable expected infectiousness (Figure S2). We then split the total variance in expected infectiousness over the course of the outbreak into between- and within-cohort components. We found that epidemics with larger $\mathcal{R}_0$ have larger between-cohort variation, as expected. However, this is balanced by smaller within-cohort variation (Figure 2, panel b). We can prove that the mean and variance of expected infectiousness across the entire outbreak is equal to one as the initial infected fraction tends zero, regardless of the value of $\mathcal{R}_0$ (Section S.1). Thus, the dispersion κ is also one.

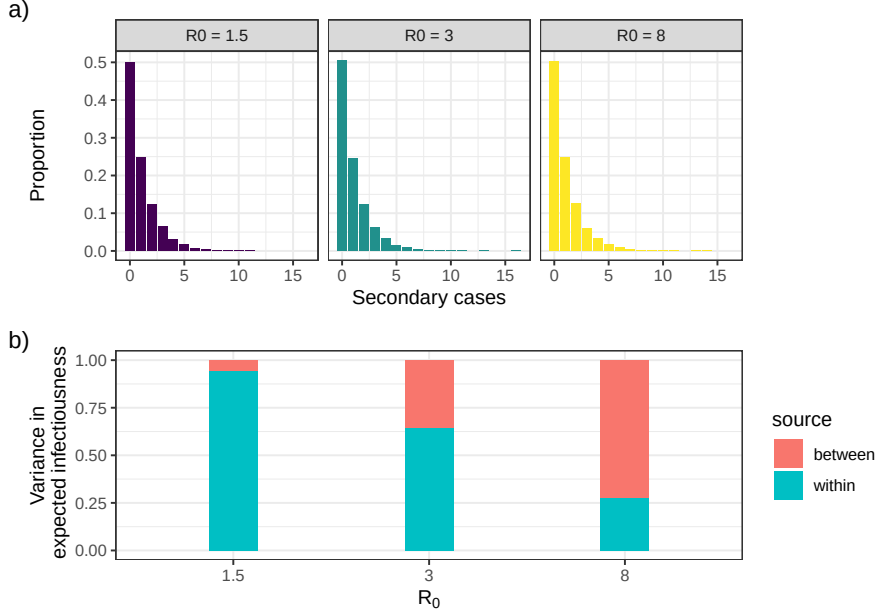


Figure 2: Heterogeneity in the number of secondary cases is independent of $\mathcal{R}_0$ when quantified across an entire outbreak. Panel a: Indistinguishable empirical secondary-case distributions for epidemics of different strengths. Epidemic outbreaks were simulated for a population of 10000, and the number of secondary cases per infector was recorded at the time of outbreak extinction. Panel b: Identical variance in expected infectiousness for epidemics of different strengths. As $\mathcal{R}_0$ increases, between-cohort variance rises, and within-cohort variance falls, maintaining constant total variance of one. In panel b, the outbreaks were modeled using standardized compartmental models.

We then investigated how expected infectiousness would evolve over time. In particular, how cohorts evolve over time and how the outbreak appears when ob-

served at specific time points. To do so, We considered three different values of $\mathcal{R}_0$, and for each value of $\mathcal{R}_0$, we simulated an outbreak using the standardized deterministic SIR model and truncated the simulation at several cutoff (time) points. At each cutoff point, we calculated heterogeneity in expected infectiousness, measured by κ , between cases generated by that time. In the early stage of the outbreak, observed heterogeneity was lower than in the linearized model or the full outbreak, and it varied by $\mathcal{R}_0$. From the beginning, in epidemics with large $\mathcal{R}_0$, the between-cohort variation was the main driver of heterogeneity. By the end of the outbreaks, epidemics with different $\mathcal{R}_0$ values eventually showed the same heterogeneity in expected infectiousness (Figure 3).

We also examined an idealized scenario in which, at each cutoff point, we included only cohorts that had become infectious by that time. We then calculated mean and heterogeneity of expected infectiousness, accounting for these cohorts' contributions to transmission throughout the entire outbreak. For smaller values of $\mathcal{R}_0$, both mean and heterogeneity of expected infectiousness among cohorts infected early were closed to those for a linearized model, i.e., a mean of $\mathcal{R}_0$ and $\kappa = 1$ (Figure 4). As the outbreak unfolds, depletion of the susceptible population reduced mean expected infectiousness. By the end of the outbreak, this mean approached one (the almost equality between the total number of secondary cases and the total number of infected individuals, provided a negligible initial cohort size relative to the final outbreak size). Regardless of $\mathcal{R}_0$, within-cohort variation primarily drove heterogeneity among early-infected cohorts. After the peak incidence, susceptible depletion reduced the share of within-cohort variation, a reduction that was more pronounced in epidemics with large $\mathcal{R}_0$. By the end of the outbreak, all epidemics with different $\mathcal{R}_0$ agreed on both mean and heterogeneity of expected infectiousness.

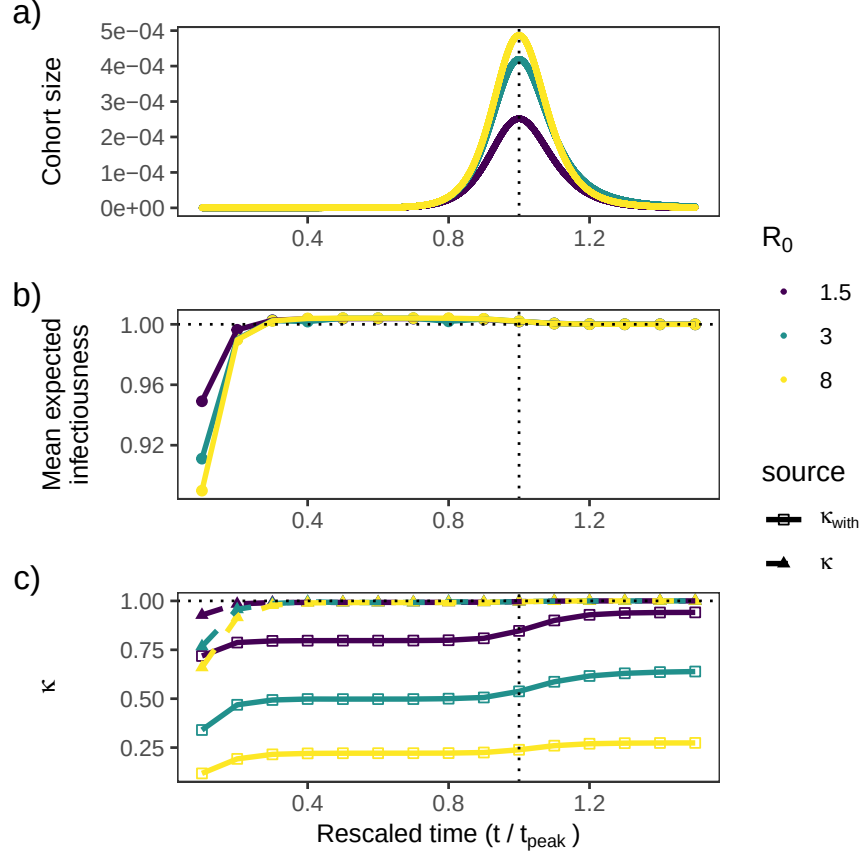


Figure 3: **Expected infectiousness evolves over the course of an outbreak.** Panel a: The size of the cohort infected during $[t, t + \Delta t]$ is given by the area under the incidence curve over that interval, where $\Delta t = 10^{-4}t_{\text{peak}}$ and t_{peak} denotes the time of peak incidence. Panel b: At each time point, mean expected infectiousness is computed using the cases realized by that time. Early in the outbreak, the initial infectious cohort constitutes a relatively large fraction of all infected population up to that time. Because the infectors of this initial cohort are not included, the mean expected infectiousness is noticeably below one. At early times ($t \lesssim 0.4t_{\text{peak}}$), epidemics with larger R_0 have accumulated fewer infections, making the effect of this omission more pronounced. Panel c: κ increases over the course of the outbreak. For epidemics with smaller R_0 , a given fraction of t_{peak} corresponds to a longer elapsed time, allowing differences in infectious periods among cohort members to become apparent. As a result, the within component of κ (measured as the ratio of within-cohort variance to the squared mean) is larger than that for epidemics with larger R_0 . For panels b and c, the simulation was truncated at each plotted time. The y-axis value at each time point was computed by assigning to each cohort the realized case reproductive number up to that time. In all panels, time units measured in the mean infectious period are rescaled relative to the peak time, so that peak incidence occurs at ($t/t_{\text{peak}} = 1$).

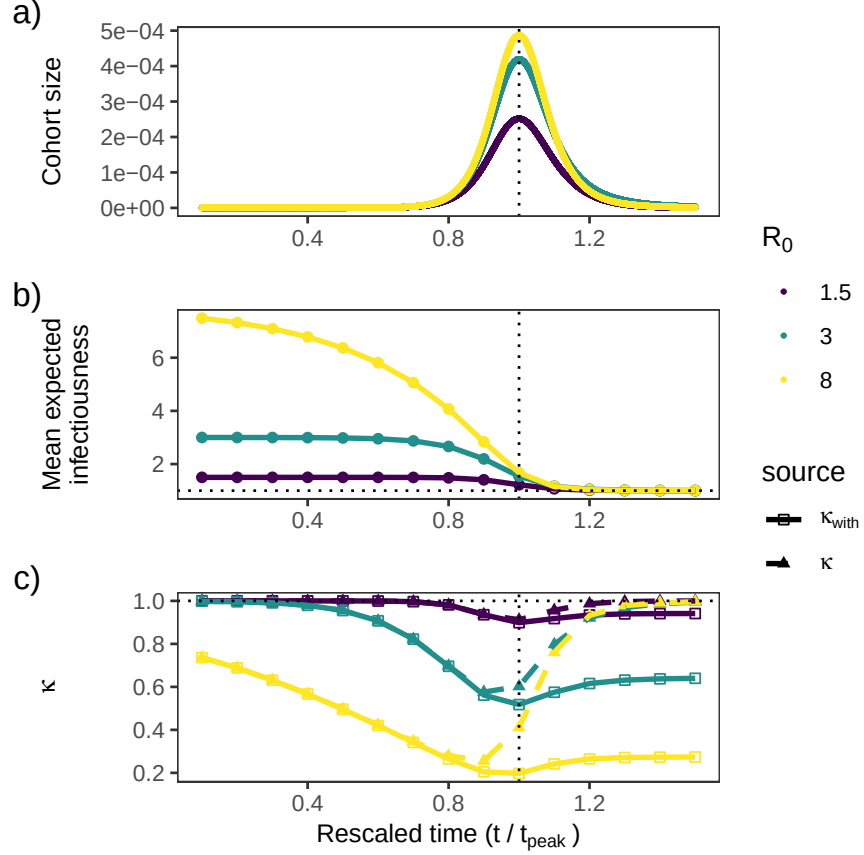


Figure 4: **Forecasts of the distribution of expected infectiousness evolve as an outbreak unfolds.** Panel a: Cohort size infected during $[t, t + \Delta t]$ is measured as the area under the incidence curve over this interval ($\Delta t = 10^{-4}t_{\text{peak}}$, where t_{peak} is the time of peak incidence). Panel b: At each time point, mean expected infectiousness is calculated across all cohorts infected by that time, including the initial infectious cohort of size 10^{-9} , while accounting for each cohort's subsequent contributions to disease transmission. Cohorts infected earlier have higher mean expected infectiousness, with values depending on key parameter R_0 . As incidence increases, later-infected cohorts, who have larger sizes, encounter susceptible pool depletion, decreasing expected infectiousness. After the peak, the susceptible population size stabilizes, and the mean approaches one. Panel c: The within-cohort component of the squared coefficient of variation κ_{with} , measured as the ratio of the within-cohort variance to the squared mean, is the main driver of heterogeneity among the early-infected cohorts, as they experience a similar size of susceptible population. As the outbreak unfolds, the decline in the susceptible population drives between-cohort variation, which is more pronounced in epidemics with large R_0 . κ reaches its minimum around the outbreak peak; large cohorts experience sharp depletion of susceptibles, and the impact of variation in recovery time fades, declining the variance, and hence κ . After the peak, κ rebounds to one. For panels b and c, the y-axis value at each time point was quantified using only the cohorts that had been infected up to that time.

3 Discussion

Quantifying heterogeneity in secondary-case distributions helps better understand how epidemics unfold and how they can be contained. Although there have been efforts to both quantitatively describe individual-level heterogeneity in contact rates [18] and to parameterize biologically realistic secondary case distributions, abstracted from the mechanism [12], these efforts have been largely separate from the construction and evaluation of the compartmental models that enable epidemic forecasting and intervention design (but see [13]). Here, we expand the bridge between individual- and population perspectives of the heterogeneity by studying epidemiologically-relevant cohorts, those infected at the same time.

To do so, we first describe a statistic, κ , for quantifying the emergent heterogeneity entailed in the SIR models and their stochastic realizations. Next, we compare the emergent heterogeneity from deterministic ODE SIR models to the dispersion in secondary case distributions resulted from their stochastic realizations. Finally, we consider how different time frames dictate different views of emergent heterogeneity, and how expected infectiousness evolves through time in epidemics of different strengths.

We observed that early in the outbreak, mean and variance of the expected infectiousness vary by $\mathcal{R}_0$, so do those of secondary-case distributions. However, when computed across the entire outbreak, these quantities become independent of $\mathcal{R}_0$. We indeed prove that, provided a negligible fraction of initial infectors, both mean and variance are equal to one, regardless of $\mathcal{R}_0$. Disentangling the within- and between-cohort variances reveals that in epidemics with large $\mathcal{R}_0$, unlike those with smaller $\mathcal{R}_0$, the between-cohort variance dominates the within-cohort variance.

In real-time analyses, κ , used as a measure of heterogeneity, remains below one.

There are some limitations to our work. The underlying model is based on a closed, well-mixed population. It is not clear how heterogeneity in mixing propensity might impact the independence of heterogeneity in expected infectiousness from $\mathcal{R}_0$ and, hence, alter κ . Working with SIR models implicitly implies an exponential infectious period. Investigating the impact of infectious period distribution on the heterogeneity in expected infectiousness is left for future work.

[TG: Can we make a note about for epidemics with large R_0 , if you don't start tracking cases right from the beginning, you'll already underestimate cases/case] **[JD: Yes, this should go into the paper.]**

In SIR-type epidemics, the effective reproduction number declines as the susceptible population becomes depleted. Consequently, if surveillance begins after the initial transmission generations, estimates of secondary-cases will represent the reduced effective reproduction number rather than $\mathcal{R}_0$ [15]. As a result, delayed surveillance may yield substantially lower estimates, especially when $\mathcal{R}_0$ is large, due to rapid depletion of the susceptible pool.

4 Materials and Methods

The values of secondary cases in panel (a) of Figure 1 were generated by computing the geometric probability mass function with mean $\mathcal{R}_0$ at points $0, 1, \dots, 30$. We used the convention that the geometric distribution counts the number of failures before the first success. As for the expected infectiousness, we evaluated the exponential probability density function with mean $\mathcal{R}_0$ at 300 equally distanced point in the interval $[0, 30]$.

To generate panel (b) of Figure 1, we followed the Lloyd-Smith's approach and computed the inequality in the activity distribution [12, SI, Sec. 2.2.5]. More specifically, we used the relation $F_{\text{trans}}(x) = \frac{1}{\mathcal{R}_0^2} \int_0^x v e^{-\frac{v}{\mathcal{R}_0}} dv$ which is the fraction of cases due to those caused up to x secondary cases. The fraction of cases due to those caused more than x cases would be $1 - F_{\text{trans}}(x)$, which is equal to $(1 + \frac{x}{\mathcal{R}_0}) e^{-\frac{x}{\mathcal{R}_0}}$. The population fraction of the individuals infected more than x cases is $e^{-\frac{x}{\mathcal{R}_0}}$. We used a similar approach to compute the inequality in the secondary case distribution: We used the relation $F_{\text{trans}}(x) = \frac{1}{\mathcal{R}_0} \sum_{v=0}^x v G(v, \mathcal{R}_0)$ which is the fraction of cases due to those caused up to x secondary cases. Here $G(v, \mathcal{R}_0)$ is a geometric distribution with mean $\mathcal{R}_0$. The fraction of cases due to those caused more than x cases equals $1 - F_{\text{trans}}(x)$. The population fraction of these individuals would be $1 - P(x)$, where $P(x)$ is the cumulative distribution function of a geometric distribution with mean $\mathcal{R}_0$ evaluated at x .

To generate panel (a) of Figure 2, we used the individual based simulation and computed the number of secondary cases in a population of size 10^4 . The simulation was initiated with one case.

We used a simple deterministic SIR model to compute the mean and variance of expected infectiousness $\mathcal{R}_c$. First, we scaled time by the mean infectious period, so the resulting SIR model then depends on one parameter, $\mathcal{R}_0$. The SIR differential equations were numerically solved for the proportions of susceptible $x(t)$ and infectious $y(t)$ at each time point t . We then used the **R** function `approxfun` to construct a function, $X(t)$, that linearly interpolates the time-series $(t, \mathcal{R}_0 x(t))$. We associated a cohort to each of the first 60% time points in the time-series t (We used a 60% cutoff to ensure cohorts had almost recovered by the end of the simulation period) and calculated the cohort-specific mean and variance.

More specifically, for the cohort infected at time point τ , the mean, $\mu(\tau)$, and variance $\sigma^2(\tau)$ of $\mathcal{R}_c$ read as:

$$\mu(\tau) = \int_{t>\tau} f(t - \tau) \int_{\tau}^t \mathcal{R}_0 x(s) ds dt, \quad (1a)$$

$$\mathbb{E}[\mathcal{R}_c^2(\tau)] = \int_{t>\tau} f(t - \tau) \left(\int_{\tau}^t \mathcal{R}_0 x(s) ds \right)^2 dt, \quad (1b)$$

$$\sigma^2(\tau) = \mathbb{E}[\mathcal{R}_c^2(\tau)] - \mu^2(\tau). \quad (1c)$$

Here $f(\cdot)$ is the probability density function of the infectious period. To calculate these quantities for the cohort infected at τ , the interpolating function $X(t)$ was

integrated from τ to t , which is the case reproductive potential associated with the cohort fraction recovered at time point t . The mean $\mathcal{R}_c$ for each cohort $\mu(\tau)$ was then the integral of the case reproductive potential weighted by the infectious period density function. We used the same approach to compute $\mathbb{E}[\mathcal{R}_c^2(\tau)]$ and variance of $\mathcal{R}_c$ for each cohort ($\sigma^2(\tau)$). In Figure S2, the middle panel (panel b) was generated using this approach.

We obtained mean expected infectiousness $\mathcal{R}_c$ by integrating the cohort-specific mean $\mu(\tau)$ against the incidence, $i(\tau) = \mathcal{R}_0 x(\tau) y(\tau)$, and normalizing the result by the final size, $\int i(t) dt$. To compute the between cohort variance, we integrated the squared cohort mean $\mu^2(\tau)$ weighted by the incidence $i(\tau)$ and divided it by the final size. The result minus the squared mean of $\mathcal{R}_c$ equaled the between cohort variance. The within-cohort variance was computed by taking integral over the cohorts' variance $\sigma^2(\tau)$ weighted by the incidence. The result was also normalized by the final size. The total variance was the sum of the between- and within-variances. We also calculated the total variance independently; we integrated $\mathbb{E}[\mathcal{R}_c^2(\tau)]$ weighted by the incidence and divided it by the final size. The result minus the squared mean of $\mathcal{R}_c$ yielded the total variance. Both approaches produced the same value for the total variance.

We used deterministic simulations to compute the y-axis values in the panels b and c of Figure 3. Specifically, at each time point t_f , we considered the cohorts infected up to time t_f , including the initial infectious cohort. We then computed cohort statistics by calculating the cases generated by these cohorts up to time t_f . For example, for a cohort infected at time point $\tau < t_f$, the mean, $\mu(\tau, t_f)$, and variance of $\mathcal{R}_c$, $\sigma^2(\tau, t_f)$, read as:

$$\mu(\tau, t_f) = \int_{t=\tau}^{t_f} f(t-\tau) dt \int_{\tau}^t \mathcal{R}_0 x(s) ds + F(t_f - \tau) \int_{\tau}^{t_f} \mathcal{R}_0 x(s) ds, \quad (2a)$$

$$\begin{aligned} \mathbb{E}[\mathcal{R}_c^2(\tau, t_f)] &= \int_{t=\tau}^{t_f} f(t-\tau) dt \left(\int_{\tau}^t \mathcal{R}_0 x(s) ds \right)^2 \\ &\quad + F(t_f - \tau) \left(\int_{\tau}^{t_f} \mathcal{R}_0 x(s) ds \right)^2, \end{aligned} \quad (2b)$$

$$\sigma^2(\tau, t_f) = \mathbb{E}[\mathcal{R}_c^2(\tau, t_f)] - \mu^2(\tau, t_f), \quad (2c)$$

where $F(\cdot)$ is the survival function associated with the infectious period density $f(\cdot)$. The mean and variance of $\mathcal{R}_c$ are then

$$\mu(t_f) = \frac{\int_0^{t_f} i(\tau) \mu(\tau, t_f) d\tau}{\int_0^{t_f} i(t) dt}, \quad (3a)$$

$$\sigma^2(t_f) = \frac{\int_0^{t_f} i(\tau) \mathbb{E}[\mathcal{R}_c^2(\tau, t_f)] d\tau}{\int_0^{t_f} i(t) dt} - \mu^2(t_f). \quad (3b)$$

The between-cohort component of κ at time point t_f is $\kappa_{\text{bet}}(t_f) = \sigma_{\text{bet}}^2(t_f) / \mu^2(t_f)$,

where

$$\sigma_{\text{bet}}^2(t_f) = \frac{\int_0^{t_f} i(\tau) \mu(\tau, t_f)^2}{\int_0^{t_f} i(t) dt} - \mu^2(t_f).$$

To generate panels (b) and (c) of Figure 4, as with the corresponding panels of Figure 3, at each time point t_f , we considered only cohorts infected by t_f , including the initial infectious cohort. Instead, we counted the total cases caused by these cohorts during the outbreak. To do so, we simply substituted $\mu(\tau)$ and $\mathbb{E}[\mathcal{R}_c^2(\tau)]$ for $\mu(\tau, t_f)$ and $\mathbb{E}[\mathcal{R}_c^2(\tau, t_f)]$ in Equation (3), where $\mu(\tau)$ and $\mathbb{E}[\mathcal{R}_c^2(\tau)]$ were provided in Equation (1).

All integrations were done using `lsoda` method with the R package `deSolve`. All simulations were carried out using R 4.5.2. Code for all numerical simulations is housed at: <https://github.com/dushoff/kappaCode>. We solved all integrals across a range of values for $\mathcal{R}_0$, using the starting values $y_0 = 10^{-9}$; $x_0 = 1 - 10^{-9}$ to represent the limiting case in which there are no exogenous cases. In building these simulations, we used a range of time step sizes, noting convergence towards known and conjectured values (e.g., epidemic final size, mean case reproduction number, variance in case reproduction number) as resolution increased.

S.1 Analytical Results

S.1.1 Preliminaries

The standard SIR model is described by the following equations:

$$\begin{aligned}\dot{x}(t) &= -\mathcal{R}_0 x(t)y(t), \\ \dot{y}(t) &= \mathcal{R}_0 x(t)y(t) - y(t),\end{aligned}$$

where $x(t)$ and $y(t)$ denote the proportion of susceptible and infectious individuals, respectively. Time has been normalized by the recovery rate. Let $i(t)$ denote the incidence $\mathcal{R}_0 x(t)y(t)$, and define the size of the epidemic as follows:

$$Z = \int_0^\infty dt i(t).$$

Let $f(\cdot)$ denote the probability density function of the infectious period. Define $F(t)$ as the corresponding survival distribution, that is

$$F(t) = 1 - \int_0^t d\tau f(\tau) = \int_t^\infty d\tau f(\tau). \quad (\text{S.1})$$

Note that $F(t)$ is the probability that an individual infected at time 0 remains infectious after time t .

We define the *expected case reproductive number* between time τ and ρ as

$$C(\tau, \rho) = \mathcal{R}_0 \int_\tau^\rho dt x(t).$$

The k^{th} weighted raw moment of C is then defined as

$$C_k = \int_0^\infty d\tau \int_\tau^\infty d\rho w(\tau, \rho) (C(\tau, \rho))^k, \quad (\text{S.2})$$

where $w(\tau, \rho) = i(\tau)f(\rho - \tau)$ is the size of the cohort that became infectious at time τ and recovered at time ρ .

S.1.2 The Statement

The excess variation of the individual case reproductive number in the standard SIR model outbreak is 1, provided that the initially infectious cohort size is negligible. Mathematically, this amounts to proving that

$$\boxed{\kappa = \frac{C_0 C_2}{C_1^2} - 1 = 1.} \quad (\text{S.3})$$

S.1.3 Proof

We will calculate C_0 , C_1 and C_2 . We can calculate C_0 and C_1 without making any assumptions about the functional form of f (we don't have to assume that the wait times are exponentially distributed). We do need F to be memoryless, i.e., obey $F(a+b) = F(a)F(b)$, to get the "right" value of C_2 . Calculating each of these moments essentially involves multiple manipulations of the limits of the integrals involved. To proceed, we assume that size of the initial infectious cohort is negligible.

From Equation S.2, we have

$$\begin{aligned} C_0 &= \int_0^\infty d\tau \int_\tau^\infty d\rho w(\tau, \rho) \\ &= \int_0^\infty d\tau \int_\tau^\infty d\rho i(\tau) f(\rho - \tau) \end{aligned}$$

Substituting $\omega + \tau$ for ρ in the inner integral, results in

$$C_0 = \int_0^\infty d\tau i(\tau) \int_0^\infty d\omega f(\omega).$$

Since f is the residence time distribution, the inner integral is simply 1. It follows that

$$\begin{aligned} C_0 &= \int_0^\infty d\tau i(\tau) \\ &= Z. \end{aligned}$$

The next step is to prove that $C_1 = Z$. From Equation S.2, we have

$$\begin{aligned} C_1 &= \int_0^\infty d\tau \int_\tau^\infty d\rho w(\tau, \rho) C(\tau, \rho) \\ &= \mathcal{R}_0 \int_0^\infty d\tau \int_\tau^\infty d\rho i(\tau) f(\rho - \tau) \int_\tau^\rho dt x(t). \end{aligned}$$

Instead of first integrating over t from τ to ρ and then over ρ from τ to ∞ , we can first integrate over ρ from t to ∞ and then integrate over t from τ to ∞ , i.e.,

$$C_1 = \mathcal{R}_0 \int_0^\infty d\tau \int_\tau^\infty dt i(\tau) x(t) \int_t^\infty d\rho f(\rho - \tau).$$

Substitute $\rho = \omega + \tau$ in the innermost integral and using Equation S.1, we get

$$\begin{aligned} C_1 &= \mathcal{R}_0 \int_0^\infty d\tau \int_\tau^\infty dt i(\tau) x(t) \int_{t-\tau}^\infty d\omega f(\omega) \\ &= \mathcal{R}_0 \int_0^\infty d\tau \int_\tau^\infty dt i(\tau) x(t) F(t - \tau). \end{aligned}$$

Note again that we have not used the fact that f is exponential, we have merely used the definition of F as an integral of f . By interchanging the order of the integrals once more, we have

$$\begin{aligned} C_1 &= \mathcal{R}_0 \int_0^\infty d\tau \int_\tau^\infty dt i(\tau)x(t) \int_{t-\tau}^\infty d\omega f(\omega) \\ &= \mathcal{R}_0 \int_0^\infty dt x(t) \int_0^t d\tau i(\tau)F(t-\tau). \end{aligned}$$

Note that the inner integrand counts the number of individuals that entered the infectious state at some time $\tau < t$ and remain infectious at time t . When we integrate over all times $\tau < t$, we simply get the number of individuals who are infectious at time t , i.e., $y(t)$. So,

$$C_1 = \mathcal{R}_0 \int_0^\infty dt x(t) \int_0^t d\tau i(\tau)F(t-\tau) \quad (\text{S.4})$$

$$= \mathcal{R}_0 \int_0^\infty dt x(t)y(t) \quad (\text{S.5})$$

$$= \int_0^\infty dt i(t) \quad (\text{S.6})$$

$$= Z. \quad (\text{S.7})$$

The next step is to show $C_2 = 2Z$. Given Equation S.2, we have

$$\begin{aligned} C_2 &= \int_0^\infty d\tau \int_\tau^\infty d\rho w(\tau, \rho) (C(\tau, \rho))^2 \\ &= \mathcal{R}_0^2 \int_0^\infty d\tau \int_\tau^\infty d\rho i(\tau)f(\rho-\tau) \left(\int_\tau^\rho dt x(t) \right)^2 \end{aligned}$$

Now we break down the square integral into a double integral

$$\begin{aligned} \left(\int_\tau^\rho x(t)dt \right)^2 &= \left(\int_\tau^\rho x(t)dt \right) \left(\int_\tau^\rho x(s)ds \right) \\ &= \left(\int_\tau^\rho x(t)dt \right) \left(\int_\tau^\rho x(s)ds \right) \\ &= \int_\tau^\rho \int_\tau^\rho dt ds x(t)x(s). \end{aligned}$$

In $t-s$ space, this is integrating $x(t)x(s)$ over a square region whose vertices are (τ, τ) , (τ, ρ) , (ρ, τ) and (ρ, ρ) . This is twice the integral over the triangle whose vertices are (τ, τ) , (τ, ρ) and (ρ, τ) . So,

$$\left(\int_\tau^\rho x(t)dt \right)^2 = 2 \int_\tau^\rho ds \int_\tau^s dt x(t)x(s)$$

Plugging this back into the expression for C_2 yields

$$C_2 = 2\mathcal{R}_0^2 \int_0^\infty d\tau \int_\tau^\infty d\rho \int_\tau^\rho ds \int_\tau^s dt i(\tau)f(\rho-\tau)x(t)x(s).$$

First, we reorder the integrals over ρ and s and change the limits appropriately

$$C_2 = 2\mathcal{R}_0^2 \int_0^\infty d\tau i(\tau) \int_\tau^\infty ds x(s) \int_s^\infty d\rho \int_\tau^s dt f(\rho - \tau)x(t).$$

Next, we reorder the integrals over ρ and t to get

$$C_2 = 2\mathcal{R}_0^2 \int_0^\infty d\tau i(\tau) \int_\tau^\infty ds x(s) \int_\tau^s dt x(t) \int_s^\infty d\rho f(\rho - \tau).$$

By substituting $\rho = \omega + \tau$ and using Equation S.1, we get

$$C_2 = 2\mathcal{R}_0^2 \int_0^\infty d\tau i(\tau) \int_\tau^\infty ds x(s) \int_\tau^s dt x(t) F(s - \tau).$$

We then reorder the integrals over τ and s

$$C_2 = 2\mathcal{R}_0^2 \int_0^\infty ds x(s) \int_0^s d\tau \int_\tau^s dt x(t) i(\tau) F(s - \tau).$$

Swapping the integrals over τ and t , results in

$$C_2 = 2\mathcal{R}_0^2 \int_0^\infty ds x(s) \int_0^s dt x(t) \int_0^t d\tau i(\tau) F(s - \tau).$$

Now, we use the fact that the survival function F is memoryless, i.e., $F(a+b) = F(a)F(b)$, and replace $F(s - \tau)$ with $F(s - t)F(t - \tau)$

$$C_2 = 2\mathcal{R}_0^2 \int_0^\infty ds x(s) \int_0^s dt x(t) \int_0^t d\tau i(\tau) F(s - t) F(t - \tau).$$

Next, we use the same argument we used to go from Equation S.4 to Equation S.5, which results in

$$\begin{aligned} C_2 &= 2\mathcal{R}_0^2 \int_0^\infty ds x(s) \int_0^s dt x(t) F(s - t) y(t) \\ &= 2\mathcal{R}_0 \int_0^\infty ds x(s) \int_0^s dt i(t) F(s - t) \\ &= 2\mathcal{R}_0 \int_0^\infty ds x(s) y(s) \\ &= 2 \int_0^\infty ds i(s) \\ &= 2Z. \end{aligned}$$

Finally, replacing C_0 , C_1 , and C_2 by Z , Z , and $2Z$, respectively, in Equation S.3, results $\kappa = 1$.

Contribution of Initial Infections

Mathematical derivation in the previous section was based on the assumption that the proportion of the population who were infectious initially was negligible. In what follows, we revisit the results while relaxing this assumption.

Let y_0 denote the proportion of the population that were infectious at time 0. The contribution of these initial infections can be incorporated into the expressions above if we change the lower limits of integration to 0^- and assume that the epidemic starts abruptly at time $t = 0$, i.e.,

$$i(t) = y_0 \delta(t), \quad t \in (-\infty, 0], \quad (\text{S.8})$$

where $\delta(t)$ denotes the Dirac delta distribution [2]. Then, we have

$$y(t) = \int_{0^-}^t d\tau \, i(\tau) F(t - \tau).$$

As shown below, the equation for the first moment C_1 , remains unchanged when the contribution of the initially infectious cohort is included, provided that the these initial infections are not counted in Z , i.e.,

$$Z = \int_{0^+}^t d\tau \, i(\tau).$$

Here 0^+ indicates that the size of the initially infected cohort is not included in Z .

However, the expression for the second moment C_2 changes to

$$C_2 = 2Z - 2y_0 \mathcal{R}_0 \int_0^\infty dt \, x(t) F(t) \quad (\text{S.9})$$

The second term in the right-hand side of Equation S.9 is twice the expected infectiousness of the initially infected cohort.

The total weight C_0 will now includes the initially infectious cohort size as well, i.e.,

$$C_0 = Z + y_0.$$

The first raw moment C_1 is given by

$$\begin{aligned} C_1 &= \int_{0^-}^\infty d\tau \int_\tau^\infty d\rho \, w(\tau, \rho) C(\tau, \rho) \\ &= \mathcal{R}_0 \int_{0^-}^\infty d\tau \int_\tau^\infty dt \, i(\tau) x(t) \int_t^\infty d\rho f(\rho - \tau) \\ &= \mathcal{R}_0 \int_{0^-}^\infty d\tau \int_\tau^\infty dt \, i(\tau) x(t) F(t - \tau) \\ &= \mathcal{R}_0 \int_0^\infty dt \, x(t) \int_{0^-}^t d\tau \, i(\tau) F(t - \tau) \\ &= \mathcal{R}_0 \int_{0^-}^\infty dt \, x(t) y(t). \end{aligned}$$

The proportion of the population who were infectious at $t \in (-\infty, 0)$ is zero, and both $x(t)$ and $y(t)$ are bounded. Thus, without loss of generality, the lower limit can be replaced with 0^+ . This gives us the last term is equal to

$$\begin{aligned} C_1 &= \mathcal{R}_0 \int_{0^+}^{\infty} dt x(t)y(t) \\ &= \int_{0^+}^{\infty} dt i(t) \\ &= Z. \end{aligned}$$

Now we move on to recalculate C_2 , while taking the initially infectious cohort into account. We have

$$\begin{aligned} C_2 &= \int_{0^-}^{\infty} d\tau \int_{\tau}^{\infty} d\rho w(\tau, \rho) (C(\tau, \rho))^2 \\ &= \mathcal{R}_0^2 \int_0^{\infty} d\tau \int_{\tau}^{\infty} d\rho i(\tau) f(\rho - \tau) \left(\int_{\tau}^{\rho} dt x(t) \right)^2 \\ &= 2\mathcal{R}_0^2 \int_{0^-}^{\infty} ds x(s) \int_{0^-}^s dt x(t) \int_{0^-}^t d\tau i(\tau) F(s-t) F(t-\tau) \\ &= 2\mathcal{R}_0^2 \int_{0^-}^{\infty} ds x(s) \int_{0^-}^s dt x(t) F(s-t) y(t) \end{aligned}$$

We split the innermost integral into two integrals

$$\begin{aligned} C_2 &= 2\mathcal{R}_0^2 \int_{0^-}^{\infty} ds x(s) \int_{0^-}^s dt x(t) F(s-t) y(t) \\ &= 2\mathcal{R}_0^2 \int_{0^-}^{\infty} ds x(s) \left(\underbrace{\int_{0^-}^{0^+} dt x(t) F(s-t) y(t)}_0 + \int_{0^+}^s dt x(t) F(s-t) y(t) \right) \\ &= 2\mathcal{R}_0^2 \int_{0^-}^{\infty} ds x(s) \int_0^s dt x(t) y(t) F(s-t) \\ &= 2\mathcal{R}_0 \int_{0^-}^{\infty} ds x(s) \int_0^s dt i(t) F(s-t) \end{aligned}$$

The innermost integral gives the proportion of the population who entered the infectious state at some time $t > 0$ and remain infectious at time s . Hence, it equals the total proportion infectious at time s minus the proportion of the population who were initially infectious and remain infectious at time s . Then

$$\begin{aligned} C_2 &= 2\mathcal{R}_0 \int_{0^-}^{\infty} ds x(s) (y(s) - y_0 F(s)) \\ &= 2\mathcal{R}_0 \int_{0^-}^{\infty} ds x(s) y(s) - 2\mathcal{R}_0 \int_{0^-}^{\infty} ds x(s) y_0 F(s) \\ &= 2Z - 2\mathcal{R}_0 \int_0^{\infty} ds x(s) y_0 F(s) \end{aligned}$$

Note that changing the lower limit of integration from 0^- to 0 does not affect the result because the integrand is bounded.

The moments of expected infectiousness at some point during the outbreak

In this section, we derive the moments of C at an arbitrary time $j \in [0, \infty)$ during the outbreak, while accounting for the initially infectious cohort. Up to time j , the expected infectiousness of individuals who became infectious at time $0 < \tau < j$ and remain infectious at time j is the equal to that of the individuals who became infectious at τ and recovered at time j . To capture this observation, we introduce a modified infectious period density function, denoted $g_j(t)$. This function equals the true infectious period density function $f(t)$ for $t < j$, and assigns the remaining probability mass to $t = j$. Hence, its cumulative distribution function reaches 1 at j , i.e.,

$$g_j(t) = \begin{cases} f(t), & \text{if } t < j, \\ F(j)\delta(t - j), & \text{otherwise.} \end{cases}$$

Accordingly, we redefine the size of the cohort that became infectious at time τ and recovered at time ρ as follows

$$w_j(\tau, \rho) = i(\tau)g_j(\rho - \tau), \quad \tau \leq \rho \leq j.$$

Note that $\int_t^j g_j(s) ds$ is equal to $F(t)$ for $t \leq j$. It can be shown that the total weight, $C_0(j)$ equals

$$\int_{0^-}^j dt i(t).$$

The first raw moment $C_1(j)$ is given by

$$\begin{aligned} C_1(j) &= \int_{0^-}^j d\tau \int_{\tau}^j d\rho w_j(\tau, \rho) C(\tau, \rho) \\ &= \mathcal{R}_0 \int_{0^-}^j d\tau \int_{\tau}^j d\rho i(\tau) g_j(\rho - \tau) \int_{\tau}^{\rho} dt x(t) \\ &= \mathcal{R}_0 \int_{0^-}^j d\tau \int_{\tau}^j dt i(\tau) x(t) \int_t^j d\rho g_j(\rho - \tau) \\ &= \mathcal{R}_0 \int_{0^-}^j d\tau \int_{\tau}^j dt i(\tau) x(t) F(t - \tau) \\ &= \mathcal{R}_0 \int_0^j dt x(t) \int_{0^-}^t d\tau i(\tau) F(t - \tau) \\ &= \mathcal{R}_0 \int_{0^-}^j dt x(t) y(t) \\ &= \int_{0^+}^j dt i(t). \end{aligned}$$

Now we move on to calculate $C_2(j)$. We have

$$\begin{aligned}
C_2(j) &= \int_{0^-}^j d\tau \int_{\tau}^j d\rho w_j(\tau, \rho) (C(\tau, \rho))^2 \\
&= \mathcal{R}_0^2 \int_{0^-}^j d\tau \int_{\tau}^j d\rho i(\tau) g_j(\rho - \tau) \left(\int_{\tau}^{\rho} dt x(t) \right)^2 \\
&= 2\mathcal{R}_0^2 \int_{0^-}^j ds x(s) \int_{0^-}^s dt x(t) \int_{0^-}^t d\tau i(\tau) F(s - t) F(t - \tau) \\
&= 2\mathcal{R}_0^2 \int_{0^-}^j ds x(s) \int_{0^-}^s dt x(t) F(s - t) y(t) \\
&= 2\mathcal{R}_0^2 \int_{0^-}^j ds x(s) \int_0^s dt x(t) y(t) F(s - t) \\
&= 2\mathcal{R}_0 \int_{0^-}^j ds x(s) \int_0^s dt i(t) F(s - t) \\
&= 2\mathcal{R}_0 \int_{0^-}^j ds x(s) (y(s) - y_0 F(s)) \\
&= 2\mathcal{R}_0 \int_{0^-}^j ds x(s) y(s) - 2\mathcal{R}_0 \int_{0^-}^j ds x(s) y_0 F(s) \\
&= 2 \int_{0^+}^j i(s) ds - 2\mathcal{R}_0 \int_0^j ds x(s) y_0 F(s)
\end{aligned}$$

Supplementary Figures

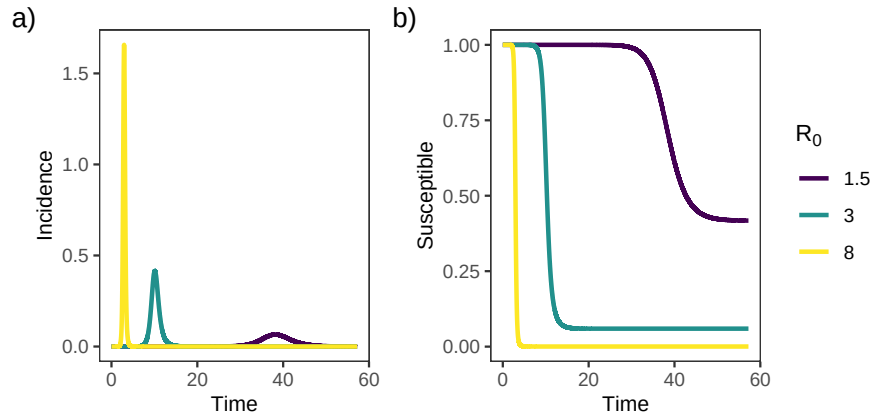


Figure S1: Outbreaks with higher R_0 values lead to greater variation in incidence and more rapid depletion of the susceptible pool. We modeled these outbreaks using a deterministic compartmental approach where time is in units of mean infectious period.

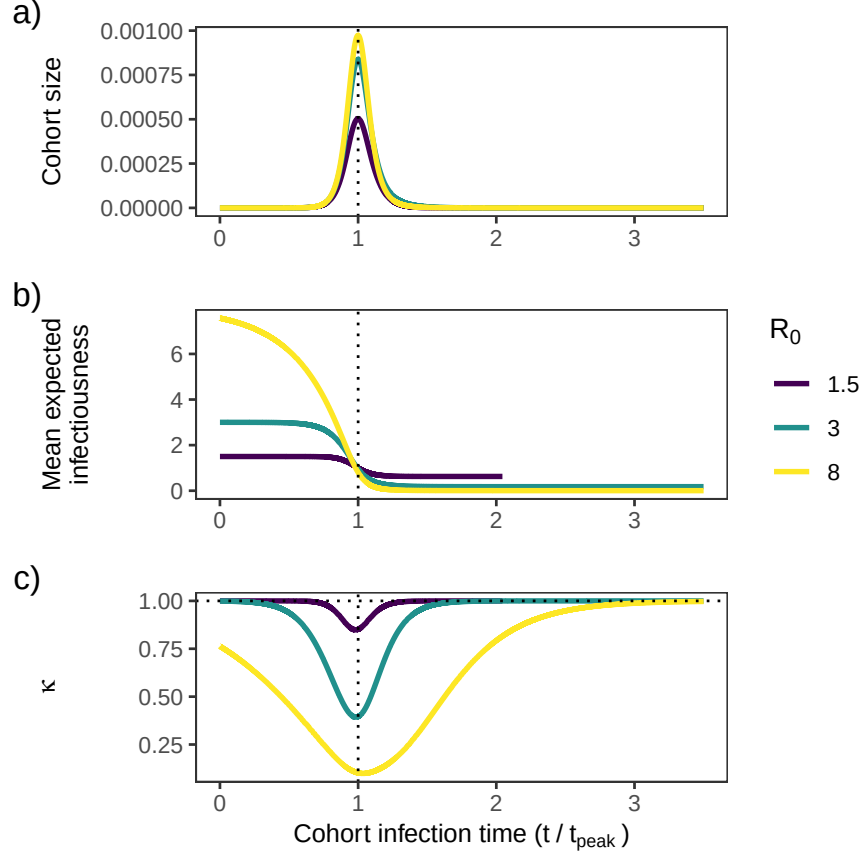


Figure S2: **Expected infectiousness of cohorts.** Panel a depicts cohort size as a function of the time of infection, in units of the mean infectious period divided by the incidence peak time. Individuals infected during $[t, t + \Delta t]$ form a cohort, with $\Delta t = 2 \times 10^{-4}$. In Panel b, mean expected infectiousness associated with each cohort is plotted. Early in the outbreak, the susceptible pool shrinks slowly, and differences in recovery time mainly cause variations in case numbers. As a result, the average for cases is about the same as in a simple linearized model. Near the outbreak's peak, faster susceptible loss partly compensates for the inequality stemming from the infectious period, and hence, the squared coefficient of variation κ decreases (Panel c). After the peak, once the susceptible pool has become nearly constant again, variation in the infectious period once more drives variation within cohorts; κ approaches one. In stronger outbreaks (higher R_0), it takes longer (in rescaled time units) for the susceptible population to stabilize, explaining the slower move of κ toward one.

References

- [1] Benjamin M Althouse, Edward A Wenger, Joel C Miller, Samuel V Scarpino, Antoine Allard, Laurent Hébert-Dufresne, and Hao Hu. Super-spreading events in the transmission dynamics of sars-cov-2: Opportunities for interventions and control. *PLoS biology*, 18(11):e3000897, 2020.
- [2] David Champredon, Jonathan Dushoff, and David J. D. Earn. Equivalence of the erlang-distributed seir epidemic model and the renewal equation. *SIAM Journal on Applied Mathematics*, 78(6):3258–3278, 2018.
- [3] Paul Z Chen, Niklas Bobrovitz, Zahra Premji, Marion Koopmans, David N Fisman, and Frank X Gu. Heterogeneity in transmissibility and shedding sars-cov-2 via droplets and aerosols. *Elife*, 10:e65774, 2021.
- [4] Akira Endo, Sam Abbott, Adam J Kucharski, Sebastian Funk, et al. Estimating the overdispersion in covid-19 transmission using outbreak sizes outside china. *Wellcome open research*, 5:67, 2020.
- [5] T Garske and C J Rhodes. The effect of superspreading on epidemic outbreak size distributions. *J Theor Biol*, 253:228–37, 2008.
- [6] Ashish Goyal, Daniel B Reeves, and Joshua T Schiffer. Multi-scale modelling reveals that early super-spreader events are a likely contributor to novel variant predominance. *J R Soc Interface*, 19:20210811, 2022.
- [7] Lea Hamner. High sars-cov-2 attack rate following exposure at a choir practice—skagit county, washington, march 2020. *MMWR. Morbidity and mortality weekly report*, 69, 2020.
- [8] Herbert W Hethcote. The mathematics of infectious diseases. *SIAM Review*, 42(4):599–653, 2000.
- [9] Alex James, Jonathan W Pitchford, and Michael J Plank. An event-based model of superspreading in epidemics. *Proceedings of the Royal Society B: Biological Sciences*, 274(1610):741–747, 2007.
- [10] W. O. Kermack and A. G. McKendrick. A contribution to the mathematical theory of epidemics. *Proceedings of the Royal Society A.*, 115(772):700–721, 1927.
- [11] Jacob E Lemieux, Katherine J Siddle, Bennett M Shaw, Christine Loreth, Stephen F Schaffner, Adrienne Gladden-Young, Gordon Adams, Timelia Fink, Christopher H Tomkins-Tinch, Lydia A Krasilnikova, et al. Phylogenetic analysis of sars-cov-2 in boston highlights the impact of superspreading events. *Science*, 371(6529):eabe3261, 2021.
- [12] J O Lloyd-Smith, S J Schreiber, P E Kopp, and W M Getz. Superspreading and the effect of individual variation on disease emergence. *Nature*, 438:355–9, 2005.

- [13] Michael T Meehan, Angus Hughes, Romain R Ragonnet, Adeshina I Adekunle, James M Trauer, Pavithra Jayasundara, Emma S McBryde, and Alec S Henderson. Replicating superspreader dynamics with compartmental models. *Sci Rep*, 13:15319, 2023.
- [14] Bjarke Frost Nielsen, Kim Sneppen, and Lone Simonsen. The counter-intuitive implications of superspreading diseases. *Nat Commun*, 14:6954, 2023.
- [15] Lorenzo Pellis, Paul J. Birrell, Joshua Blake, Christopher E. Overton, Francesca Scarabel, Helena B. Stage, Ellen Brooks-Pollock, Leon Danon, Ian Hall, Thomas A. House, Matt J. Keeling, Jonathan M. Read, JUNIPER Consortium, and Daniela De Angelis. Estimation of reproduction numbers in real time: Conceptual and statistical challenges. *Journal of the Royal Statistical Society Series A: Statistics in Society*, 185:S112–S130, 11 2022.
- [16] D Schenzle. An age-structured model of pre- and post-vaccination measles transmission. *IMA J Math Appl Med Biol*, 1:169–91, 1984.
- [17] Kim Sneppen, Bjarke Frost Nielsen, Robert J Taylor, and Lone Simonsen. Overdispersion in COVID-19 increases the effectiveness of limiting nonrepetitive contacts for transmission control. *Proc Natl Acad Sci U S A*, 118, 2021.
- [18] M E Woolhouse, C Dye, J F Etard, T Smith, J D Charlwood, G P Garnett, P Hagan, J L Hii, P D Ndhlovu, R J Quinnell, C H Watts, S K Chandiwana, and R M Anderson. Heterogeneities in the transmission of infectious agents: implications for the design of control programs. *Proc Natl Acad Sci U S A*, 94:338–42, 1997.